

A Theory of Economic Slack

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CHAPTER 5.

Prevalence of price and wage rigidity

In a slackish market, there are infinitely many ways to split the bilateral surplus enjoyed by traders who have met, and therefore many possible ways to set prices. We assume that it is price norms that determine prices in the market.

Price norms are determined by both economic and social forces and can take many forms. Okun (1981, p. 105) explains this wonderfully in the case of wages—and the same applies to prices:

Wage standards are conventional and habitual. They have their logic and take on a life of their own. They become entrenched because firms have incentives to get into step with other participants in the labor market. There may be more determinacy in the persistence of any reasonable convention than in its precise nature.

Because price norms can take many forms, before modelling them we must examine how prices behave in real-world markets to understand their properties. Accordingly, in this chapter, we review evidence on prices and wages in the United States. This review will show that prices and wages are somewhat rigid: they adjust slowly to shocks that disturb the market. Fairness appears to be a key reason behind price and wage rigidity. In chapter 6, we will model a realistic price norm and plug into our slackish market model.

5.1. What are price norms?

A price norm is just how buyers and sellers set prices in a market. There is no such thing in a Walrasian market because there the price is set to equalize supply and demand. Sellers have no choice but to adhere to the market price. If a seller raises their price 1 cent above the market price, they will lose all customers. Buyers have no choice but to adhere to the market price. If a buyer asks for a discounted price 1 cent below the market price, the seller will just refuse to trade and sell their goods to other buyers. And of course there is no reason for sellers to lower their price below the market price or for buyers to offer a price above the market price since sellers and buyers can already trade the desired quantity at the market price.

But assuming a price norm is not as strange as it might seem at first. In the real world, there are also many ways in which prices are formed—a lot of it being cultural or institutional. In the service industry in the United States, for instance, prices are jointly determined by buyers and sellers. Sellers set a floor price, and then buyers add a tip on top, at their discretion in theory, but following well-established tipping norms in practice. A tremendous amount of goods and services are transacted that way, where cultural norms determine the price. Between 2008 and 2018 in the United States, tips declared on W-2 forms amounted to \$30 billion per year (Basker, Foster, and Stinson 2024). Furthermore tips are widely underreported: only between 45% and 60% of tips are reported. If underreporting is 50%, the W-2 numbers imply yearly tips of \$60 billion—offered entirely voluntarily by buyers after consumption.

Tips are an example of how culture influences prices. The minimum wage and other price controls show how institutions might influence prices. Huet-Vaughn and Piqueras (2023) provide fascinating evidence from a minimum wage reform in Iowa in 2015, which affected wages paid by firms for many years although the reform itself only lasted a bit more than a year. In 2015–2016 in a county in Iowa, the minimum wage was raised from \$7.25 an hour to \$10.10 an hour. The reform obviously raised the wages of minimum-wage workers. But the minimum-wage reform was abolished after 17 months by the state of Iowa, bringing the minimum wage in the county back down to \$7.25. Despite the abolition, wages remained elevated for minimum-wage workers at least until 2022. Firms voluntarily chose to pay workers better for many years, influenced by a temporary policy intervention.

Price norms determine price levels, but as Okun (1981, pp. 151–154) observed, they also include various practices to build the confidence of customers that prices are as fair and as predictable as possible. For instance, firms might commit to fixed-time scheduling: to hold price constant for a fixed amount of time. They might for instance commit to adjust prices only once a year or once a season. Firms might also pre-notify customers in advance of any price change, and explain why an upcoming price increase is necessary.

5.2. Why do price norms exist?

A key property of slackish markets is that buyers and sellers are always happy to trade. Anytime a seller and a buyer find each other and there is an exchange, there is a bilateral surplus from the trade so that both buyer and seller take something away from it—they are strictly better off trading than not trading. This surplus offers room for price norms to emerge and exist.

This is true in the real world: sellers are usually happy when they can sell their goods while buyers are happy when they can buy what they were looking for. This is different from the Walrasian model, where the marginal buyer and seller are indifferent between buying or selling and not buying or not selling, because the price exactly equals the seller's marginal cost of production and the buyer's marginal utility.

Since the solution to such a bilateral-monopoly problem is indeterminate, there are many possible ways to split the bilateral surplus between seller and buyer, including norms.¹ Assuming a price norm is unusual in economics models, but the reality is that in a slackish model, there is no obvious economic mechanism that can determine the price. Any price that leaves seller and buyer with a positive individual surplus would be acceptable to both parties. There is no deviation from that price that generates a Pareto improvement. Of course, a seller would be better-off with a higher price, but a buyer would be worse-off. In this setting, market forces cannot erode prevailing price norms.

In fact, such theoretical indeterminacy might explain why there exist so many different cultural and institutional arrangements to determine prices—effectively norms. In some places, sellers simply set a fixed price. In other places, people normally bargain over the prices of goods. There are places where a price is posted but customers are expected to add a tip whose size is dictated by complex conventions. Sometimes, prices are regulated or directly set by governments.

The situation of bilateral monopoly arises because the pairing of a buyer and a seller generates a positive bilateral surplus. Intuitively, the reason is that it is difficult for sellers to sell their goods and for buyers to find a good, so there is a surplus when they meet and have the opportunity to trade. We now formally establish what the bilateral surplus is. We consider the ex-post surplus from trade, evaluated once the buyer has incurred the cost to visit the seller and the seller has incurred the cost to produce the good.² The outside option is to walk away after sinking those costs. To start, we compute the seller's and buyer's surpluses: what they take away from the trade of a good, compared with what they would take away if the trade did not occur.

¹The indeterminacy of the solution to the bilateral-monopoly problem has been known since Edgeworth (1881). It is discussed by Howitt and McAfee (1987) and Hall (2005) in the context of matching models.

²In this chapter production is exogenous and costless—each seller just brings k_i to the market—but in chapter 7 production will be endogenous and costly.

5.2.1. Seller surplus

We assume that just like buyers, sellers have a quasilinear utility function, so their marginal utility of money is 1. The price of a transaction is p . Then, for p units of money, the seller takes away p utils from the trade. If there was no trade, the seller would get nothing. The seller does not benefit from the good that they put for sale because they cannot consume their own production. The seller surplus therefore simply is:

$$\mathcal{S} = p.$$

Critically, the seller surplus is positive for any positive price, so the seller is always happy to trade.

5.2.2. Buyer surplus

Given that a buyer's utility is given by (4.10), and that we evaluate the surplus at the buyer's planned consumption level c , the marginal utility of one good is:

$$\frac{\partial \mathcal{U}}{\partial c} = (1 - \alpha)ac^{-\alpha}.$$

This is what the buyer takes away from the trade of an extra good at the planned consumption level. If the trade did not occur, however, the buyer would get to keep their p units of money, which provide p utils since the utility is quasilinear. Of course, the buyer does not recover the already-sunk matching expenditures that brought them here. Thus, the buyer surplus is

$$(5.1) \quad \mathcal{B} = (1 - \alpha)ac^{-\alpha} - p.$$

However, the buyer's consumption is not random. It is given by the solution to the buyer's optimization problem. It turns out that consumption is such that the buyer's surplus is always positive. The first-order condition from the buyer's problem, given by (4.13), tells us that the marginal utility from consumption, $(1 - \alpha)ac^{-\alpha}$, equals $[1 + \tau(\theta)]p$, where $\tau(\theta)$ is the matching wedge. Combining this result with the expression for the buyer surplus, we simply get:

$$(5.2) \quad \mathcal{B} = \tau(\theta)p.$$

Similarly to the seller surplus, the buyer surplus is positive. The buyer surplus has a simple interpretation: it is the monetary cost sunk to match with this seller, since consuming one good requires purchasing $\tau(\theta)$ goods for matching beforehand, each costing a price p . Another interpretation is that the buyer surplus is the monetary cost

required to match with a different seller if the present trade fell through. Indeed, matching with a new seller to purchase one good would require to spend again on matching cost. The implication is that the buyer surplus is larger when matching costs are larger, because a larger matching cost implies a larger matching wedge, τ . In any case, given that the buyer decides to shop fully knowing what the price norm is, when they find a good, their surplus is always positive, so they are always happy to buy it.

5.2.3. Total surplus

Since the buyer and seller surpluses are both positive, we conclude that the total surplus from a trade, $\mathcal{T} = \mathcal{S} + \mathcal{B}$, is also positive. The total surplus admits two simple, equivalent expressions. The first expression involves the price of the good and is obtained by using (5.2):

$$(5.3) \quad \mathcal{T} = [1 + \tau(\theta)] p.$$

The second expression involves the marginal utility of consumption and is obtained by using (5.1):

$$(5.4) \quad \mathcal{T} = (1 - \alpha)ac^{-\alpha}.$$

Hence we see that the total surplus is simply the marginal utility of consumption for buyers. This is quite obvious in retrospect. Without trade, the good is not sold and therefore not consumed by anyone, so one unit of consumption is lost, which is valued at the buyer's marginal utility. In the slackish model, therefore, all trades that occur generate a positive total surplus, irrespective of the price norm.

In sum, any trade generates a positive surplus for both the buyer and the seller. This is a key distinction from Walrasian models where marginal traders are indifferent. The model also differs from monopolistic models, in which sellers enjoy a surplus but the marginal buyer is indifferent between buying and not buying. And the model differs from monopsonistic models, in which buyers enjoy a surplus but the marginal seller is indifferent between selling and not selling. With price norms giving all bargaining power to either seller or buyer, however, the slackish model can mimic a monopolistic or monopsonistic model. But it is much more general than these models: it can feature any price between the monopolistic and monopsonistic extremes—in general, both traders derive some surplus from any trade.

5.2.4. Bilateral efficiency

The theoretical reason why any price norm may prevail in reality is that any price norm respects bilateral efficiency. Price norms that do not respect bilateral efficiency would impose unquestionable losses on traders and would therefore be too costly to be sustainable. It therefore seems that at the minimum, to be sustainable, a price norm must respect bilateral efficiency.

Bilateral efficiency is just pairwise rationality: a minimal amount of cooperation so that when two traders meet, they do the best they can for the pair as a whole. If a pair has the opportunity to trade for a positive total surplus, they do it. If bilateral efficiency was violated, pairs would forego opportunities to generate a positive joint surplus and would in effect lower their joint well-being. They would forego available improvements to their joint well-being. If there was minimal cooperation, pairs would not ignore these Pareto improvements and would take the opportunity to increase their total surplus. Hence, price norms under minimal cooperation should be bilaterally efficient.

To respect bilateral efficiency, a price must be at such level that when a trade generates a positive total surplus, buyer and seller are also left with positive individual surpluses. In that way, any time a trade creates some bilateral gains it also creates individual gains, which makes it acceptable to both trading parties. This ensures that any trade that creates bilateral gains is conducted. We have just established that for any price norm, all trades conducted on the market generate both individual and total surpluses. Therefore, any price norm respects bilateral efficiency.³

5.2.5. Barro critique

Starting with the *General Theory*, many models have used wage rigidity to generate unemployment. Barro (1977) offered a famous critique of such wage rigidity. He noted that wage rigidity generated unemployment by forcing firms to fire workers whose current marginal product of labor fell below the real wage. However, Barro noted that such firing violated bilateral efficiency: firms and workers could both be better off and generate a positive total surplus by renegotiating a lower wage and continuing to work together. Barro added that since workers are employed by firms for a long time, such renegotiations might be likely to occur. He concluded that generating unemployment from such rigidity was therefore unsatisfactory. The price and wage norms used in our slackish model respect

³A famous result due to Hall (2005) is that in a DMP model, fixed wages are theoretically legitimate as long as they remain within a certain band. Why is the price norm not restricted to a price band here? The reason is that Hall works with a linear model, whereas our utility function satisfies the Inada condition: it has infinite marginal utility when consumption falls to 0. This ensures that any price is acceptable to buyers when their planned consumption is low enough. Given the buyers take the price norm as given when they decide to shop, they only plan to purchase goods that deliver at least as much utility as the price they pay for it, so the buyer surplus is always positive. Given the Inada condition, there is no restriction on the price norm: the price band becomes infinitely wide when planned consumption becomes arbitrarily low.

bilateral efficiency—all relationships generating a positive surplus are preserved—and are therefore immune to the Barro critique.

5.3. Evidence of price rigidity

In theory, price norms can take many forms. We therefore need to examine how prices are set in the real world to understand how price norms look in reality.

In this section, we argue that US prices are somewhat rigid. What do we mean when we say that prices are rigid? We mean that prices do not fully respond to underlying shocks. An extreme case of this is when prices do not move at all—in that case the price is fixed. This is the opposite of a flexible price, which fully responds to shocks and absorbs them.

5.3.1. Frequency of price changes

A typical piece of evidence in favor of price rigidity is that prices stay at the same level for a number of months. Nakamura and Steinsson (2013, table 1) report the frequency of price changes and duration for which prices stayed the same in the United States across many different products. The statistics come from the Consumer Price Index microdata, collected by the BLS between 1988 and 2005.

For regular prices, which exclude sales prices, the median durations are about 7–10 months, while the mean durations are a bit higher, around 9–12 months. The mean is bigger than the median because some products have prices that stay stable for a very long time. A classic example is the bottle of Coca-Cola, which remained priced at a nickel (5 cents) for more than 70 years, from 1886 until 1959 (Levy and Young 2004).

The noticeable difference between median and mean durations of price spells tells us that there is quite a lot of heterogeneity in the frequency of price changes across items. We see this clearly on the histogram reported by Nakamura and Steinsson (2013, figure 3). Many items have a frequency of regular-price changes that is very close to 0, meaning that prices are fixed over years. The biggest mass is for products that have a price-change frequency of 5% per month. These products' prices are fixed on average for $1/0.05 = 20$ months, so about a year and a half. For most of the products, the probability of a price change is below 10% per month; these products' prices are fixed on average for at least $1/0.1 = 10$ months, so close to a year. The rest of the products are scattered all over the frequency range. There is clearly a lot of heterogeneity in how prices change, which might be unsurprising, because both pricing norms and the variability of demand and production costs vary across products.

5.3.2. Prevalence of rigid prices

In slackish markets, goods are traded only once buyers and sellers have matched. The price is determined in a situation of bilateral monopoly, in which both buyer and seller derive a surplus from trading. Critically, the total surplus is determined by the buyers' valuation for the good (equation (5.4)). As a result, the central empirical question is how prices respond to shifts in how buyers value the good. Unfortunately, the frequency of price changes does not tell us how prices respond to shocks. If the demand for a good is not changing, there is no reason for its price to change. We therefore need to turn to different evidence.

In effect, we are interested in the passthrough of valuation shocks into prices. The idea would be to determine how much the price of a good changes if customers suddenly value it differently. A flexible price would adjust one-for-one, a rigid price would adjust some but less than one-for-one, and a fixed price would not respond.

Unfortunately, again, estimates of the passthrough of marginal valuation into prices are not readily available. But there is plenty of other evidence that firms are reluctant to adjust prices in response to demand shocks. Based on copious econometric evidence, Nordhaus (1974, pp. 183–184) reported that:

Faced with temporary changes in demand, firms generally alter production and employment rather than price. Prices are based on long-run profitability and other managerial objectives and are not significantly adjusted to cyclical conditions.

Okun (1975, p. 362) observed in discussions with business people that “empirically, the typical standard of fairness involves cost-oriented pricing with a markup,” implying that firms would not think about responding to a shift in demand. Blinder et al. (1998, p. 156) asked 121 US firms the following: “When the demand for your product is weak, are your customers willing to let you hold your prices, or do they insist on price reductions?” If the price norm was flexible, all customers would insist on price reductions when they value the product less, but that is not what happened in practice: only 29% of firms answered that customers insist on price reductions; 51% of firms answered that they can hold their prices; and the rest said that customers have mixed attitudes.

The total surplus from a trade can also be expressed as a function of market tightness (equation (5.3)). This is because market tightness determines the matching wedge—the amount of resources that buyers must devote to matching with a seller. When it is difficult to find sellers, then having found one is particularly valuable. If we take this perspective, then we might expect prices to respond directly to market tightness. But here too, survey evidence suggests that prices are quite rigid. Among 189 US firms interviewed by Blinder et al. (1998, p. 153), 64% agree with the idea that “firms have implicit understandings with

their customers—who expect the firms not to take advantage of the situation by raising prices when the market is tight.”

5.3.3. Possible reasons for price rigidity

We now briefly try to explain why prices might be rigid: why might sellers not fully respond to demand shocks? In a Walrasian market, the price goes up whenever demand goes up, so why are sellers reluctant to charge a higher price for their goods if buyers value them more?

The main reason for price rigidity is that people care about the fairness of prices and react negatively when they perceive prices as unfair. As a result, sellers avoid charging prices that customers might consider unfair. In a famous survey, Kahneman, Knetsch, and Thaler (1986, p. 729) document that buyers find it unfair for sellers to raise prices when the demand for their good is high. They describe the following situation:

A hardware store has been selling snow shovels for \$15. The morning after a large snowstorm, the store raises the price to \$20.

Among 107 respondents, only 18% regard this pricing as acceptable, whereas 82% regard it as unfair.

We saw in chapter 4 that if demand for a good is higher but the market price does not respond, market tightness rises, making it more difficult for buyers to buy the good. One might wonder whether, in this situation, it would become acceptable for sellers to raise prices. Kahneman, Knetsch, and Thaler (1986, p. 732) actually inquire about this situation:

A shortage has developed for a popular model of automobile, and customers must now wait two months for delivery. A dealer has been selling these cars at list price. Now the dealer prices this model at \$200 above list price.

Among 130 respondents, only 29% regard this pricing as acceptable, whereas 71% regard it as unfair. So buyers also find it unfair for sellers to raise prices when the market for their good is tight.

We also saw in chapter 4 that market tightness rises if supply of a good is lower and the market price does not respond. One might wonder in this situation too whether it would become acceptable to raise prices. Kahneman, Knetsch, and Thaler (1986, p. 734) inquire about this question by describing the following vignette:

A severe shortage of Red Delicious apples has developed in a community and none of the grocery stores or produce markets have any of this type of apple on their shelves. Other varieties of apples are plentiful in all of the stores. One grocer receives a single shipment of Red Delicious apples at the regular

wholesale cost and raises the retail price of these Red Delicious apples by 25% over the regular price.

Among 102 respondents, only 37% view this pricing as acceptable, whereas 63% view it as unfair. This vignette confirms that buyers find it unfair for sellers to raise prices when market tightness is high.

Kahneman, Knetsch, and Thaler conducted their survey in Toronto and Vancouver, but subsequent studies have confirmed their results in the United States. For instance, Shiller, Boycko, and Korobov (1991) obtained similar results from a survey conducted in the greater New York area in 1990.

A natural question is: why should firms care about what is fair and what is not fair? This question has a simple answer: customers appear to reduce purchases when they feel unfairly treated (Piron and Fernandez 1995). More generally, unfair prices seem to make customers angry, which of course firms would like to avoid (Rotemberg 2009).

Moreover, firms do identify fairness as a major concern in price setting. Blinder et al. (1998, tables 5.1 and 5.2) finds that it is in part because of “implicit contracts” with customers that firms do not respond to shocks more: “firms tacitly agree to stabilize prices, perhaps out of fairness to customers.” In fact, US firms report that the main reason why they do not change prices more often is that “it would antagonize or cause difficulties for our customers” Blinder (1994, table 4.5). And when firms must change prices, they devote significant resources to justify the price increase to customers and avoid antagonizing them, especially when the increase is large (Zbaracki et al. 2004).

5.4. Evidence of wage rigidity

In this section, we turn from prices to wages. We review evidence on how wages are set in the real world, both for workers in established relationships with firms and for new hires. The main takeaway is that wages are somewhat rigid in reality, which means that they respond in a muted fashion to shocks to labor productivity and labor market slack.

Studying wages separately from prices is important because wages are prices in the labor market model of chapter 10. Moreover, in our core business-cycle model of chapter 13, workers are self-employed, so the product and labor markets are blended, which means that prices are also wages. In that context, understanding wage rigidity is crucial to understand price rigidity. Finally, in the extension of the business-cycle model with separate product and labor markets, covered in chapter 15, both price and wage rigidity play a crucial role.

5.4.1. Frequency of wage changes for job stayers

Just as we did with prices, we begin by looking at the frequency of wage changes in the United States. Dickens et al. (2007, figure 1) report the distribution of percentage changes of wages received by workers who stayed in the same job in the United States in 1987. The data come from the Panel Study of Income Dynamics (PSID), a large-scale household survey run by the University of Michigan.

The first interesting observation is that there is a substantial spike at zero: more than 15% of workers do not have a wage change between 1986 and 1987; they have a wage freeze or a fixed nominal wage. This is surprising since the United States was experiencing inflation during that time. Therefore, real wages for these workers actually fell from one year to the next.

Another interesting point is that a significant mass of wage changes is around 4%. In that period, US inflation was around 4%, so this is evidence of real wages being fixed: nominal wages adjusted exactly by the amount of inflation.

The presence of nominal and real wage freezes are typical in the PSID and other household surveys. These freezes imply that for many workers, their nominal or real wage does not change from year to year: it remains fixed for several years. Kahn (1997) finds such patterns in the PSID between 1970 and 1988. Card and Hyslop (1997) find the same patterns in CPS microdata for 1979–1993.

Interestingly, there are a significant number of workers whose nominal wages decrease. Since Keynes (1936), macroeconomic models often assume that nominal wages are downwardly rigid—that nominal wages cannot be reduced. However, we do see nominal wage reductions in the data. Kahn (1997, table 1) reports that a nominal wage cut from one year to the next occurs 18% of the time. Card and Hyslop (1997, p. 75) find that 15%–20% of job stayers report nominal wage cuts in any year.

One possible issue with household surveys is measurement error, which might explain the nominal wage cuts that we see in the data (Akerlof, Dickens, and Perry 1996). Lebow, Saks, and Wilson (2003, figure 3) address this issue by using the BLS Employment Cost Index microdata for 1981–1999, which are based on establishment records and therefore less prone to measurement error. They find the same patterns: nominal and real wage freezes are common, but nominal wage cuts do occur. Elsby and Solon (2019, table 1) report an even more convincing piece of evidence on the frequency of downward nominal wage rigidity: in administrative data covering all workers from the US state of Washington, 2005–2015, between 20% and 33% of workers receive year-to-year nominal wage cuts, depending on the year. Finally, Grigsby, Hurst, and Yildirmaz (2021, pp. 430–431) use administrative data collected by a large payroll processing company (ADP), covering millions of US workers between 2008 and 2016. They report that on average the share of workers experiencing an annual cut in nominal base wage is only 2.5%, but this share grows

to 6% during the Great Recession, and to 10% during the Great Recession in industries particularly badly affected by the downturn. The average share of workers experiencing an annual reduction in nominal hourly compensation (including bonus and overtime) is 17%.

We have seen that many job stayers keep the same wage from one year to the next. So what is the expected duration of wages in jobs? Barattieri, Basu, and Gottschalk (2014, table 6) look at the Survey of Income and Program Participation, which is administered by the US Census Bureau, and which they correct for measurement error. They find that between 1996 and 2000, the probability of a within-job nominal wage change is 15%–22% per quarter. The implied duration of the nominal wage is between $1/0.22 = 4.5$ quarters and $1/0.15 = 6.7$ quarters, so between one year and one year and a half. Grigsby, Hurst, and Yildirmaz (2021, table 3) report similar results: a quarterly probability of nominal base wage change of 19%, which implies a mean duration of wage contracts of $1/0.19 = 5.3$ quarters.

Overall, as Altonji and Devereux (2000, p. 383) write:

The data overwhelmingly rejects a model of flexible wage changes and provides some evidence against a model of perfect downward rigidity in favor of a more general model.

Based on the evidence, and unlike in Keynesian models, this book does not rely on downward nominal wage rigidity to generate unemployment and unemployment fluctuations. Instead, we will use real wages that are somewhat rigid both upward and downward.

5.4.2. Frequency of wage changes for new hires

We have seen that within a firm-worker relationship, wages are not moving freely. However, when we consider dynamic labor market models, the key decision that the firm makes is whether they hire a worker or not, and what matters to them at the time of making that decision is how they expect to pay the worker during the entire duration of their relationship. Thus, when a worker is hired, the fact that their wages are going to be fixed over time is not sufficient information for the firm. What the firm needs to know is the total wage bill expected over the entire relationship. And what affects the firm's decision to post vacancies and hire workers is whether the overall wage payments are rigid or not.

This is not to say that the wages of existing workers are irrelevant, but they are only one piece of the puzzle. We need to know what happens to the wages of new hires. Once we combine these two bits of information, we will have the full picture of the wage that is paid during the entire relationship. And we will then build a model around that.

To know what happens to wages of new hires, we need to look at wages that are attached to vacancies that are posted. This will tell us what happens when a new worker

comes into a firm. Hazell and Taska (2025) do exactly this. They use data from a business analytics company that collects vacancies of many online job portals and company websites (Burning Glass). The data show how the wage associated with each job vacancy in a given firm and location varies between 2010 and 2020 in the US labor market. They find significant rigidities in wages that are posted with vacancies. They report the quarterly probability of nominal posted wage change for one job within a specific firm in a specific location is only 17%, and the expected duration for which the wage remained at the same level is 5.4 quarters—about a year and a half (Hazell and Taska 2025, table 3). This is in line with what we saw earlier for job stayers: many workers do not get a change in nominal wage from year to year.

When wages do change, Hazell and Taska (2025, table 3) also look at whether they go up or down. They find that there is a 4% chance of nominal wages decreasing, and a 12% chance of them increasing. This means that $4/16 = 25\%$ of wage changes are decreases, and 75% are increases. It is not surprising that increases are more common, especially as the economy was recovering from the Great Recession: technological progress, inflation, and the tightening of the economy all pushed wages up. The 25% of nominal wage decreases is also in line with the fraction of job stayers experiencing cuts in nominal wages.

Last, Hazell and Taska (2025, figure 3) look at the distribution of wage changes. It is only for non-zero wage growth, meaning it eliminates the mass of vacancies for which wages do not change from quarter to quarter. There are some nominal wage decreases, and many more wage increases. A large share of the nominal wage increases amount to 3%, and occur every six quarters. The peak of the distribution therefore implies that the typical posted wage grows at 2% per year, which is roughly the level of inflation in the United States during that time. This shows that fixed real wages are common for job vacancies.

5.4.3. Prevalence of rigid wages

We have seen that nominal and real wages move sluggishly, both in existing employment relationships and new employment relationships. However, when we talk about wage rigidity, we are thinking about how wages respond to shocks. Real wages are rigid if they do not respond fully to a change in productivity, and flexible if they respond one-for-one to a change in productivity.

In fact, wage rigidity to productivity shocks is isomorphic to price rigidity to demand shocks. In a slackish market, wages and prices are not set to clear markets but to split the bilateral surplus generated by a match. A change in labor productivity shifts the marginal value of a match to the firm, just as a change in preferences shifts the marginal value of a match to the buyer in the goods market. In both cases, the price paid in the relationship would adjust one-for-one to the underlying shock under flexible prices or wages, and less

than one-for-one under rigid prices or wages.

Haefke, Sonntag, and van Rens (2013) look at how the real wage paid to US workers changes when there is a change in that worker's productivity, combining several BLS datasets. They look at what happens to wages in both new and existing employment relationships, painting a complete picture of what happens to real wages over the entire relationship.

Over the 1984–2006 period, the elasticity of real wages to productivity is 0.24–0.37 for all workers and 0.79–0.83 for new hires (Haefke, Sonntag, and van Rens 2013, table 4). However, data are available for a longer period, 1979–2006. Over that longer period, wages are more rigid: the elasticity of real wages to productivity is 0.18–0.20 for all workers and 0.30–0.49 for new hires (Haefke, Sonntag, and van Rens 2013, table 9). The elasticities of real wages with respect to productivity below 1 are clear evidence of wage rigidity. At the same time, wages of job stayers and new hires respond to changes in productivity, and therefore real wages are not completely fixed.

Overall, the elasticity of new hires' real wages with respect to productivity is in the 0.3–0.8 range. A reasonable estimate for the elasticity is 0.5, as it is in the middle of the range and corresponds to the most compelling estimate produced by Haefke, Sonntag, and van Rens (2013, table 9): the elasticity of hourly wages over the complete data sample. Haefke, Sonntag, and van Rens (2013, p. 898) also find that wages in ongoing contracts are close to a random walk, meaning that in expectation the wage in an entire employment relationship is determined by the wage at the beginning of the relationship. The elasticity of the present value of wages with respect to labor productivity is therefore close to the elasticity of the wages of newly hired workers: about 0.5.

As we discussed in the case of prices, the total surplus from a trade can also be expressed as a function of market tightness, because market tightness determines the amount of resources that buyers must devote to matching with a seller. In the case of the labor market, this means that the total surplus from matching is lower when the labor market is slack and recruiting is easy; one wonders in this situation whether it would become acceptable to lower wages. But here too, evidence suggests that wages are quite rigid. A waves of in-migration into a labor market raises the labor force, which reduces market tightness if wages do not fall, following the mechanism described in chapter 4. As the surplus from firm-worker matches is lower, wages might fall.

However, migration does not generally seem to significantly affect local wages. Card (1990) examines the impact of international immigration on wages in a famous natural experiment: the arrival of Cuban immigrants from the Mariel boatlift on the Miami labor market in the 1980s. The well-known findings by (Card 1990, p. 255) are that “the Mariel immigration had essentially no effect on the wages . . . of non-Cuban workers in the Miami labor market ” and “perhaps even more surprising, the Mariel immigration had no strong

effect on the wages of other Cuban.” The lack of response of wages is especially surprising because the wave of immigration was substantial: Miami’s labor force increased by 7% due to the Mariel boatlift.

Boustan, Fishback, and Kantor (2010) reach a similar conclusion by studying the effect of internal migration in a quasi experiment that occurred during the Great Depression in the United States. They use variation in the generosity of New Deal programs and extreme weather events to instrument for migrant flows to and from US cities. They find that in-migration had little effect on the hourly earnings of local workers.

Maybe even more surprisingly, policies that directly affect the surplus from worker-firm matches do not seem to influence wages. Take unemployment insurance: unemployment benefits provide income to job seekers, which improves their outside option, and according to bargaining theory should raise wages. But that is not what we observe: in the United States, unemployment benefits do not seem to affect wages. Marinescu (2017) explores data on job vacancies and job applications from a major job board (CareerBuilder.com). She looks at the extensions of unemployment insurance implemented in the United States between 2008 and 2011, which raised benefit duration from 26 weeks up to 99 weeks. She finds that the benefit extensions did not affect the level or distribution of wages advertised by firms (Marinescu 2017, table 6). Johnston and Mas (2018) study in administrative data the effects of a natural experiment: an unexpected 16-week reduction in benefit duration in Missouri in 2011. They find that re-employment earnings do not change after a cut in the duration of UI benefits Johnston and Mas (2018, p. 2501).

5.4.4. Possible reasons for wage rigidity

We have seen that wages are quite rigid and do not adjust like flexible wages would. We now briefly explain why that is.

To understand wage rigidity, it helps to look at the history of the US labor market. In the early 20th century, American firms shifted from a spot-market system—where workers were hired and paid on a daily basis—to a system of internal labor markets that produced rigid wages (Jacoby 1984). Internal labor markets were more bureaucratic, rule-based, and rigid than spot markets. Wages were tied to job descriptions rather than to how productive an individual worker happened to be on a given day or how bad the labor market was that month. The motivation for the organization of firms around internal labor markets was to provide a fair, equitable remuneration to workers—to improve their job satisfaction and eventually their productivity and loyalty to the firm.

In fact, this change in how US firms pay workers might have started with industrialization at the end of the 19th century. Hanes (1993, p. 751) reports that wages became more rigid at that time as workers started to organize and did not hesitate to strike: “Having

learned that cutting nominal wages in a downturn might provoke a strike, firms avoided doing so even though none was bound by an enforceable wage contract.”

Internal labor markets might explain why, during the Great Depression, job seekers offered to work at wages much below pre-depression wages for their skill and experience in their occupation, and yet firms chose to pay existing and new workers wages that were much closer to pre-depression wages (Simon 2001). Today, internal labor markets are prevalent and they render wages insensitive to marginal productivity or labor market slack (Doeringer and Piore 1971). Having access to 20 years of personnel data from a large US firm, Baker, Gibbs, and Holmstrom (1994, p. 923) find evidence of an internal labor market: the firm “seems to be shielding its employees from some of the market-induced variations in marginal product.”

Internal labor markets also impose an internal fairness constraint: new hires and incumbent workers ought to be treated equitably. Hence, these institutions ensure that wages and wage rigidity are comparable for new and incumbent workers in similar jobs, as Grigsby, Hurst, and Yildirmaz (2021, p. 464) find.

Managers themselves have now completely internalized the ideas that led to the adoption of internal labor markets. They are convinced that cutting pay in bad times is a bad idea because it is seen as unfair by workers. Detailed interviews of compensation managers by Bewley (1999) in hundreds of US firms show that they avoid wage cuts at all costs, even in recessions, because they think wage cuts will antagonize workers and damage morale. Low morale in turn has a range of negative consequences for firms: workers get demotivated or angry, they stop trying as hard, turnover goes up, hiring gets harder, and eventually profitability drops. These findings are confirmed by other interviews and surveys of US firms (Blinder and Choi 1990; Campbell and Kamlani 1997).

And experimental evidence supports managers’ views. In 1914, Ford decided to double pay for workers by introducing a five-dollar-a-day minimum wage. Raff and Summers (1987, tables 7 and 8) report that workers’ attachment to the firm increased significantly: the yearly turnover rate fell from 370% in 1913 down to 16% in 1915, and the daily absenteeism rate fell from 10% in 1913 down to 2.5% in 1915. Simultaneously, the factory’s productivity and profitability increased noticeably. While Ford’s motivations for the wage increase are unclear, he was very pleased with the results, as Raff and Summers (1987, p. S59) recount:

There was ... no charity in any way involved.... We wanted to pay these wages so that the business would be on a lasting foundation. We were building for the future. A low wage business is always insecure... The payment of five dollars a day for an eight hour day was one of the finest cost cutting moves we ever made.

More recent evidence is consistent with what happened at the Ford factory more than a century ago. In a natural experiment involving New Jersey police departments, Mas

(2006) finds that when police officers are paid less than what they think is a fair wage, their effort drops—leading to lower arrest rates, lower sentence length, and more numerous crime reports. And in numerous laboratory experiments, researchers have found that people do not just work for money; they very much care about fairness, and when they feel shortchanged by firms, performance suffers (Fehr, Goette, and Zehnder 2009). By running experiments in firms, organizational psychologists have also found that fair pay leads to better morale and loyalty to the firm (Bewley 2005).

Labor market institutions can play a role, too, in explaining wage rigidity. During the Great Depression, for example, the National Industrial Recovery Act is often blamed for keeping wages higher than they otherwise would have been, even while the economy was collapsing (Temin 1990). Another institution that plays an important role in explaining wage rigidity in more recent times, at least for low-wage workers, is the minimum wage. Indeed, the minimum wage compresses the bottom of the wage distribution, especially for female workers, which necessarily generates rigidities (DiNardo, Fortin, and Lemieux 1996, figure 1). Unions might also be a source of wage rigidity, as they raise and stabilize wages of their members. As Freeman and Medoff (1984, p. 54) explain:

The wages of union workers tend to be less sensitive to business cycle ups and downs than the wages of nonunion workers; this is true in part ... because unions are less likely to accept real wage reductions in recessions unless the senior employees' jobs are imperiled. By contrast, nonunion firms are more likely to alter wages in the short run, although nonunion wages are also not extremely sensitive to the business cycle.

However, the impact of unions on overall wages may be more limited today since union coverage in the United States has shrunk to 10% of the workforce as of 2023 (Jaeger, Naidu, and Schoefer 2025, p. 253).

Overall, internal labor markets, managerial norms, and labor market institutions all form barriers that prevent wages from adjusting flexibly to product market or labor market conditions. Firms do not tie wages tightly to productivity and treat new and existing workers alike within their internal labor market, and managers believe cuts are poison for morale.

5.5. Summary

The evidence from US markets reveals that prices and wages exhibit substantial rigidity. This rigidity stems primarily from fairness concerns. Customers find price increases due to a higher product demand or a tighter product market to be unfair. Similarly, workers find wage cuts due to lower labor demand or a slacker labor market to be unfair. Accordingly, producers keep prices stable to maintain customers' goodwill, just as firms keep wages

stable to preserve worker morale. In the slackish model, the price norm ought to feature such rigidity to be realistic.

Although the evidence presented in this chapter goes against the Walrasian orthodoxy, it is in a way completely pedestrian. It has been known for a long time that the assumption that price and wage are perfectly flexible is unrealistic and that prices and wages are rigid in reality. Hicks (1946, p. 265) for instance observed:

This assumption [that prices are perfectly flexible] must now be dropped, for it is of course highly unrealistic. In most communities there are a large number of prices which, for one reason or another, are fairly insensitive to economic forces.... This rigidity may be due to legislative control, or to monopolistic action ... It may be due to lingering notions of a "just price". The most important class of prices subject to such rigidities are wage-rates; they are affected by rigidity from all three causes. They are particularly likely to be affected by ethical notions, since the wage-contract is very much a personal contract, and will only proceed smoothly if it is regarded as "fair" by both parties.

Besides making it obvious that prices and wages are somewhat rigid, Hicks (1946, p. 266) anticipates one of the main themes of this book—that we should look at unemployment alongside other forms of slack:

Mr. Keynes goes so far as to make the rigidity of wage-rates the corner-stone of his system.... It seems to me that the more fundamental sociological implications are brought out better if we treat rigid wage-rates as merely one sort of rigid prices. It is hard to exaggerate the immediate practical importance of the unemployment of labour, but its bearing on the nature of capitalism comes out better if we look at it alongside the unemployment (and even the misemployment) of other things.

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